

Project Define Requirements and Function

Project: Study Schedule Generator

Adviser: Ang Mengchhuong

I. Functional Requirements

1. User Input Form

- Allow users to input:
 - Subjects or topics to study
 - Free days and available hours
- Provide validation for incomplete or incorrect input.

2. Schedule Generator

- Generate a personalized study schedule based on user input.
- Ensure optimal time allocation for all subjects.
- Pause Feature:
 - Allow users to pause the schedule.
 - Ensure users cannot pause for more than half of the study duration. Otherwise, the course will restart.

3. Schedule Generator History

- Store all previously generated schedules:
 - Allow users to view old schedules.
 - Display key details like the creation date, subjects, and total study hours.
- Add filtering options for quick access to specific schedules (e.g., based on date or subject).
- Provide a "Compare Schedules" feature to analyze changes or improvements over time.

4. Calendar View

- Display the generated schedule in a clear, interactive calendar format.
- Support both daily and weekly views.

5. Quiz Module

- Generate quizzes based on topics covered in the schedule.
- Include adaptive difficulty logic:
 - Adjust question difficulty based on previous performance.
- Provide immediate feedback on quiz answers, with explanations for incorrect responses.
- Store quiz scores and show progress charts over time.

6. AI Integration

- Implement AI to optimize study roadmaps:
 - Analyze user input to recommend study strategies.
 - Adjust schedules dynamically if new subjects or constraints are added.

7. User Authentication

- Allow users to create accounts or log in.
- Enable data persistence for saved schedules and progress.

8. Notifications

- Send reminders for upcoming study sessions.
- Notify users if they are falling behind or skipping sessions.

9. Progress Tracking

- Display user performance metrics, including:
 - Study hours completed.
 - Quiz scores and progress charts.

II. Technology

1. Frontend

- **Framework:** React.js
- **Key Libraries:** Material-UI (or TailwindCSS for styling), FullCalendar.js (for calendar view).

2. Backend

- Framework: Laravel
- Features:
 - RESTful APIs for schedule generation and quiz management.
 - AI integration using Python or PHP-based libraries.

3. Database

- MySQL:
 - Tables for users, schedules, subjects, quiz questions, and quiz results.

4. Hosting

- Shared hosting platform (e.g., Hostinger) to deploy both backend and frontend

5. Development Tool

- **Code Editor:** Visual Studio Code
- **Version Control:** GitHub for collaboration and version tracking.

III. Color Scheme

1. Primary Colors (Main Branding Colors)

- **#2D9CDB (Blue)** – Used for buttons, highlights, and interactive elements.
- **#27AE60 (Green)** – Represents progress, success messages, and completed tasks.

2. Secondary Colors (For Contrast & Usability)

- **#F2994A (Orange)** – Warnings, alerts, and upcoming deadlines.
- **#EB5757 (Red)** – Errors, failed quiz attempts, or missed study sessions.

3. Background & Neutral Colors

- **#F5F5F5 (Light Gray)** – Text color for readability.
- **#FFFFFF (White)** – Text color for readability.
- **#333333 (Dark Gray/Black)** – Clean background for a distraction-free UI.

Project Screen Page

I. Landing Page (Home Page)

- Introducing the application and guiding users to key features.
- Hero section with a tagline (e.g., "Optimize Your Study Schedule with AI-Powered Precision!").
- Footer with links to About, Privacy Policy, and Contact Us.

II. Register

1. By Using Gmail

2. Input Information

- Full Name
- Username
- Email
- Password

III. Login

1. Login Account

- Username
- Password

2. Forgot Password

- Email
- Code
- New Password

IV. Schedule Generator

User need to input many information to get the schedule

- Subjects or topics to study.
- Free days and available hours.

V. Schedule

1. Calendar

Card of schedule (Name, Subject, ...) End of card have button of pause

- Mini Subject Roadmap
 - Lesson Title
 - Lesson Description
 - Date (ex: 28.01.2024)
 - Time (7:00PM – 8:45PM)
 - Quiz

2. History Schedule

Show All History of Schedule

- Mini Subject Roadmap
 - Lesson Title
 - Lesson Description
 - Date (ex: 28.01.2024)
 - Time (7:00PM – 8:45PM)
 - Quiz

VI. Notification

Alert Notification:

- Time to study

- Congratulations on the success schedule
- Remind user course will be restart
- ...

VII. Setting

- 1. Edit Profile**
- 2. Contact**
- 3. Privacy and Policy**
- 4. Logout**